

# MSc Biostatistics & Health Data Science

## Missing Data Analysis - Assignment 2

The file **data.csv** contains data from a random sample of 10000 individuals from a study investigating the association of Serum albumin levels (*alb*; in g/dl) with Alanine transaminase (*alt*; in units/L), gamma-glutamyl transferase (*ggt20*; binary, with 1 if higher than 20 units/L and 0 else) and C-reactive protein (*crp*; in mg/dl). Also, we have available the gender, the age and the Body mass index (categorized) of the participants.

Complete the following tasks:

**A.** Calculate and present appropriate descriptive statistics for all the variables in the dataset. Also, count the missing values for all the variables.

**B.** Consider the estimation of the mean Alanine levels in our sample. Calculate the complete-case estimate. Then:

1. Fit the simple linear regression model of Alanine to Age, and manually fill in the missing values on Alanine based on the predicted values of this model plus a random error term drawn from  $N(0, \hat{\sigma}^2)$ , where  $\hat{\sigma}^2$  is the residual variance estimated from the fitted model. In order not to rely on a single draw, repeat (manually; write your own, simple, R program) the same process 20 times (i.e. fill in the missing values 20 times where in each one the random error draw will be different, ending up with 20 different complete vectors of Alanine levels and hence with 20 different complete estimators of the quantity of interest). Using Rubin's rules, (manually) combine the estimates from the 20 vectors. As a measure of variability for your estimate, use the 2.5th and the 97.5th percentile of the sample of the estimators you obtained or (optionally) derive the Rubin's variance estimator (check theory slides).
2. Repeat the above, using a flexible regression model with a cubic spline (use function `ns()` or `bs()`) for Age with a knot at 50 years.
3. Using a simple logistic model for the missingness in Alanine with a linear effect for Age, calculate the Inverse Probability Weighting (IPW) estimator for the mean Alanine levels.
4. Repeat the above question improving the missingness model using a cubic spline for Age (use `bs()` function) with two knots at 46 and 54 years.

For questions 3 and 4, you are advised to examine the distribution of the inverse probability weights. You are also encouraged to explore more flexible models for the missingness mechanism in Alanine transaminase as a sensitivity analysis.

Compare your results between the 4 different approaches and with the complete-case analysis and comment.

**C.** Consider as the scientific model the regression of Albumin to Alt, GGT and CRP. For all the questions below, use **mice** and **jomo** packages.

1. Use all the available variables in the dataset and fill 10 imputed datasets using multiple imputation with chained equations (mice) with the predictive mean matching method (pmm) for all the variables with missing values. Use *maxit* = 15 and appropriately combine the results to fit the scientific model.
2. Repeat the above question, choosing an appropriate imputation method (of your choice, instead of pmm) to fill in the missing values with mice based on the type of each variable to be imputed.
3. Repeat the above question, using an interaction of Age with Alt when imputing GGT and CRP (use the passive imputation technique when imputing the interaction term). Present appropriate diagnostics plot to investigate the imputation procedure.
4. Use joint modelling multiple imputation and all the available variables to fill 10 imputed datasets (use 5000 and 2000 nburn and nbetween draws respectively) and appropriately combine the results to fit the scientific model.

Compare your results between the 4 different approaches and with the complete-case estimate and comment.

The deliverables for the assignment include a file with your results (including R output) and comments/interpretations, plus your R code for all questions either in a table at the end of the file or in a separate R script. Optionally, instead of the above, you can submit the result (word/pdf) of an Rmarkdown file that combines your R code with the results.